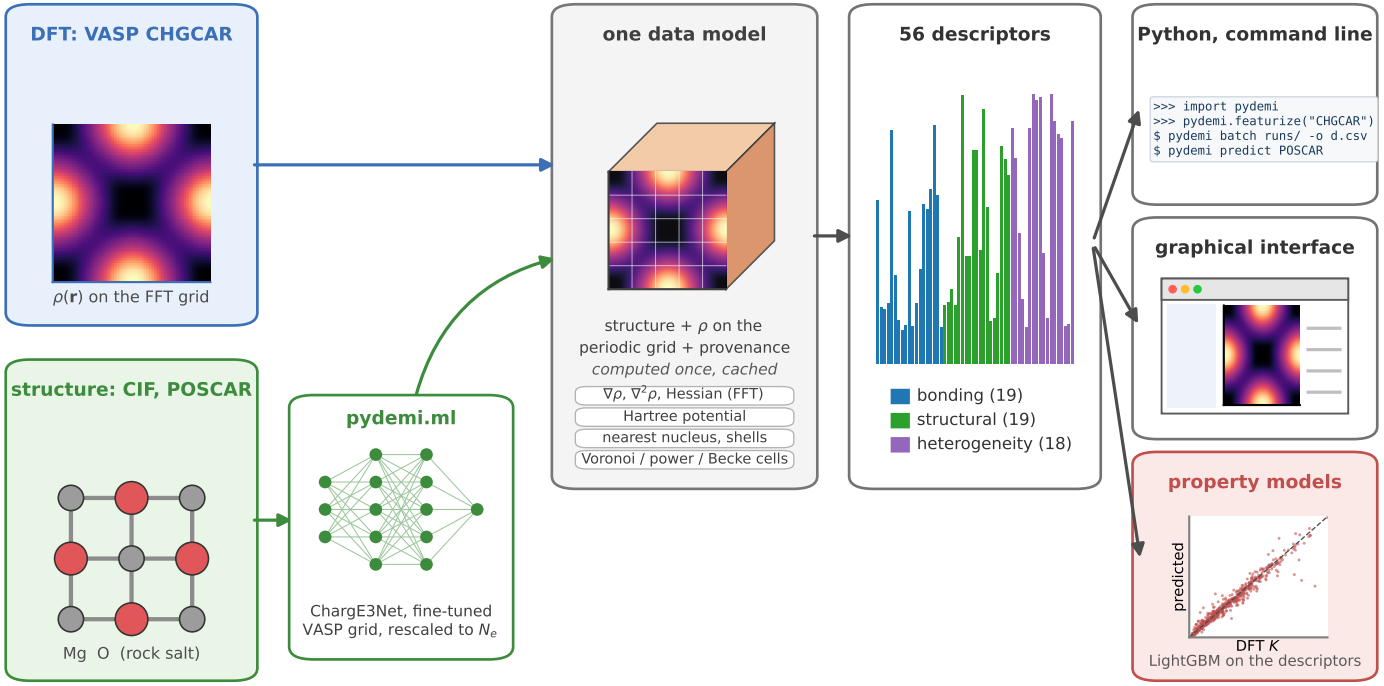

DFT: VASP CHGCAR
ρ(r) on the FFT grid
structure: CIF, POSCAR
Mg O (rock salt)
pydemi.ml
ChargE3Net, fine-tuned
VASP grid, rescaled to N_e
one data model
structure + ρ on the
periodic grid + provenance
computed once, cached
∇ρ, ∇²ρ, Hessian (FFT)
Hartree potential
nearest nucleus, shells
Voronoi / power / Becke cells
56 descriptors
bonding (19)
structural (19)
heterogeneity (18)
Python, command line
>>> import pydemi
>>> pydemi.featurize("CHGCAR")
$ pydemi batch runs/ -o d.csv
$ pydemi predict POSCAR
graphical interface
property models
predicted
DFT K
LightGBM on the descriptors